Gradient Decoupled Learning With Unimodal Regularization for Multimodal Remote Sensing Classification

Shicai Wei, Chunbo Luo, *Senior Member, IEEE*, Xiaoguang Ma, and Yang Luo

***Abstract*— The joint use of multisource remote-sensing data for Earth observation has drawn much attention due to its robust performance. Although many methods have been proposed to fuse multimodal data, they tend to improve the interaction of different modality data while ignoring the optimization of each modality. Existing studies show that high-performance modalities will suppress the learning of weak ones, leading to under-optimized multimodal learning. To this end, we propose a general framework called gradient decoupled network (GDNet) to assist the multimodal remote sensing (RS) classification. GDNet guides each modality encoder in the multimodal model to learn probabilistic representations instead of deterministic ones. This helps decouple their gradient, reducing their influence on each other and encouraging them to learn the modality-specific information. Then, we further introduce the unimodal regularization for each modality encoder to align their logit output with the multimodal one and label distribution simultaneously. This helps introduce independent gradient paths for each morality encoder to accelerate their optimization when preserving the modality-share information. Finally, extensive experiments conducted on three benchmark datasets demonstrate that the proposed GDNet can effectively address the under-optimized problem in multimodal RS image classification. Code is available at https://github.com/shicaiwei123/TGRS-GDNet.**

***Index Terms*— Classification, decoupling learning, deep learning, multimodal, remote sensing (RS).**

I. Introduction

ACCURATE land-use and land-cover classification plays an important role in many applications such as urban planning and change detection. With the rapid development of Earth observation technologies, various remote sensing (RS) resources have been widely used in RS image classification tasks. These include the light detection and ranging (LiDAR) or digital surface model (DSM) providing the height information about the ground elevation, synthetic aperture radar (SAR) providing the structure information about the Earth's surface, and hyperspectral (HS) data providing detailed content information of sensed materials. The judicious integration of these diverse data sources exploits their complementary advantages, thereby significantly enhancing the intelligent interpretation of RS data. However, the growing diversity of data also brings challenges in representing and fusing the information from different sources.

To date, numerous methods have been proposed to handle multimodal RS classification. They can be roughly divided into three categories: shallow-based methods, CNN-based methods, and Transformer-based methods. Specifically, shallow-based methods involve handcraft features like morphological profiles (MPs) [1] and extinction profiles [2], which are then classified using shallow models such as support vector machines [3] and random forests [4]. Consequently, their representation and fitting ability are usually insufficient and they thus encounter the performance bottleneck when the training data becomes complex. To this end, CNN-based methods introduce the deep convolutional neural network to promote feature extraction, fusion, and recognition for multimodal RS classification [5], [6], [7], [8], [9], [10]. Compared with the traditional methods that extract features based on expert knowledge, they perform adaptive feature extraction via the data-driven method. As a result, they can achieve powerful feature representation ability. Besides, they also improve the feature fusion via cross-modality reconstruction [11] or attention mechanism [12], enabling the model to handle the disparity between different modality data. However, their performance is still insufficient due to their limitation to capture long-range dependence [12]. To address this challenge, researchers proposed Transformer-based methods to learn and fuse the multimodal input via self-attention and cross-attention mechanism, respectively [13], [14], [15], [16]. Unlike the convolution operation that processes local neighborhoods, they compute the response as a weighted sum of the features at all positions, enabling the model to capture global information.

Despite the remarkable improvements achieved by the aforementioned methods, there are critical issues that still need to be addressed for more accurate multimodal RS classification. Existing studies show that most multimodal methods struggle to fully exploit all modalities, resulting in inadequate unimodal learning [17], [18]. Specifically, different modalities exhibit varying convergence rates, and the modalities with better performance will dominate the optimization progress, resulting in insufficient optimization of the weaker modalities. The similar problem also exists

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in the multimodal RS field. For example, the HS modality in the Huston2013 dataset contains 144 spectra while the light detection and ranging (LiDAR) modality in the same dataset only contains one channel. Consequently, the HS modality would perform better and suppress the optimization of the LiDAR modality. However, this problem seems being overlooked in existing work. To our best knowledge, our article is the first work to discuss the imbalance problem of multimodal learning in the RS field.

To address this issue, techniques such as teacher assistant and gradient modulation have been introduced to alleviate the imbalanced training. Teacher assistant introduces unimodal teachers for each modality encoder to enhance its representation ability [18]. Considering the computational burden introduced by the teacher assistant, researchers further proposed the gradient modulation method that increases the gradient of the weak modality to accelerate its learning [19], [20], [21]. While existing gradient modulation methods achieve promising results, they primarily focus on promoting the learning of weaker modalities while overlooking the dominant modality, which is influenced by the weak ones and also exhibits suboptimal performance compared to unimodal learning.

To this end, we propose a novel gradient decoupled network (GDNet) to assist multimodal RS classification. GDNet incorporates probabilistic embedding learning into each modality encoder, guiding it to learn a distribution rather than a fixed point for each input. This transformation converts the deterministic gradients of each modality encoder into distributions, alleviating their dependencies on the gradient passed from the shared fusion module and enabling the modality encoder to capture the modality-specific information. Thus, GDNet can help alleviate the optimization limitation of each modality, including the dominant one. In particular, we introduce the variance of the gradient distribution to determine the level of relaxation. Then, we further present the unimodal regularization to assist the training of GDNet. It first calculates the task loss for the representation after each modality encoder. This introduces independent gradient paths for each morality encoder, increasing the gradient of each modality encoder to accelerate their optimization and improving the gradient variance to enhance the relaxation of gradient coupling. Then, it aligns the logit output of each modality encoder with the multimodal one, encouraging them to capture the modality-share information. Here, the unimodal regularization part shares the parameters with the fusion and classification modules in the multimodal model, and thus introduces no extra parameters.

The major contributions of this work are summarized as follows.

1) We propose a novel GDNet to assist the multimodal RS image classification by guiding the encoder to learn a probabilistic embedding. This alleviates the gradient coupling between modality encoders, promoting the optimization of each modality encoder, including the dominant modality.
2) We present the unimodal regularization for GDNet to introduce independent gradient paths for each morality encoder and regularize their prediction distribution with the multimodal one. This helps accelerate their optimization to capture modality-specific information when preserving the modality-share information.
3) Experiments and ablation studies demonstrate the effectiveness and generalization of the proposed framework to promote multimodal RS image classification.

II. Related Work

A. Multimodal RS Classification via Shallow-Based Methods

The increasing availability of multimodal RS data has spurred extensive research into multimodal RS classification. Classic shallow models, including morphological operators and subspace learning, have found success in handling multimodal RS observations. For instance, Liao et al. [1] proposed a graph-based subspace embedding method to fuse MPs of HS and LiDAR data on manifolds. Tuia and Camps-Valls [22] fused the multimodal data by projecting them to a higher dimensional kernel space for better alignment. Yan et al. [23] further proposed the composite kernel-based subspace to fuse multimodal features via angle-based discriminant analysis. Hong et al. [24] proposed the CoSpace method to learn the multimodal representations via manifold alignment regularized representation learning and ridge regression, ensuring more effective information fusion. Furthermore, the same investigators extended the CoSpace model to $\mathcal{L}_1$ CoSpace by replacing ridge regression with sparse regression [25].

Despite their success, these methods often rely on artificially designed features and classify the fused representation using shallow algorithms such as support vector machines [3] and random forests [4]. Consequently, performance bottlenecks are frequently faced when handling complex training data due to their limitations in data fitting and representation abilities.

B. Multimodal RS Classification via CNN-Based Methods

In recent years, multimodal RS classification using CNN methods has attracted significant attention [6], [26], [27], [28]. In contrast to conventional approaches, CNN-based methods dispense with the requirement for manually crafted features based on expert knowledge, capitalizing on the robust feature representation capabilities inherent in deep neural networks. Specifically, they can be divided into two continuous processes: 1) extract features from multimodal data, such as HSI images and LiDAR images, individually and 2) fuse the extracted features and put them into a classifier to generate the classification result. Such a two-branch network facilitates feature extraction, yielding great improvement in the accuracy of classification tasks [29].

As for the feature extraction, Xu et al. [30] introduced a spectral–spatial CNN-based two-branch framework equipped with cascaded blocks to extract features from HS and LiDAR data. Because of the spatial and spectral complexity of HS images, many methods have been proposed to improve its feature extraction branch [30]. Ahmad et al. [31] enhanced the feature learning of HS image via a hybrid dense network with different kernels in each block. Chen et al. [32] introduced the

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Gabor operators for the CNN to guide it to acquire multiscale and multidirectional spatial features for HS images. Zhao et al. [9] extended it into the fractional domain, improving the feature diversity and discrimination. Roy et al. [33] introduced the 3D CNN layers for the HS branch to extract discriminative spectral–spatial feature representation.

Besides, numerous methods have been proposed to handle the disparity between different modality data to improve the classification performance [29]. Hang et al. [34] proposed to fuse the HS and LiDAR image via a shared CNN model and achieved performance gains. Zhao et al. [8] used an effective hierarchical random walk network to study the joint classification of HS images and LiDAR data. Feng et al. [35] proposed the dynamic scale hierarchical fusion network based on multiattention for HS and LiDAR data classification. Hong et al. [36] introduced the encoder–decoder network architecture to fuse the multimodal information by enforcing the fused features to reconstruct the multimodal input in turn. Furthermore, Wu et al. [5] proposed a cross-channel reconstructed convolutional neural network to learn a more compact fusion representation through cross-modal reconstruction strategies, which can exchange information with each other in a more efficient manner. Dong et al. [11] proposed a multibranch feature fusion network with self and cross-guided attention for HS and LiDAR classification. It refines the multimodal features extracted from HS and LiDAR streams and provides complementary information cross-guidance from the LiDAR stream to the HS stream, yielding more effective spectral–spatial-elevation information blending in the network. Mohla et al. [37] proposed a dual attention-based multimodal fusion network for HS and LiDAR classification. It utilizes the LiDAR-derived attention map to accentuate the spatial features of the HS image via the cross-attention mechanism.

C. Multimodal RS Classification via Transformer-Based Methods

Despite the impressive performance of CNNs, their effectiveness is constrained when dealing with long sequence attributes of spectral signatures due to their local inductive bias [38], [39]. To this end, researchers have increasingly turned their attention to Transformers, drawn by their capacity to model long-sequence correlations.

Xue et al. [13] and Roy et al. [15] made their first attempts to fuse HS and LiDAR data through modifying Transformers, respectively. Zhang et al. [14] proposed a local-based multisource feature interactor to explore HS and LiDAR features together. This structure alleviates the homogenization processing mode of the self-attention process. Zhang et al. [40] proposed the MTNet to enhance the Transformer-based classification of HS and LiDAR data by capturing the specific and shared characteristics simultaneously. Zhao et al. [16] integrated the CNN into the Transformer model to enhance the ability to capture spatial information. Furthermore, Yao et al. [12] extended the vision Transformer for multimodal land cover classification and proposed a general framework that is modality-agnostic and suitable for more practical scenarios under multimodal and cross-modal settings.

D. Under-Optimized Multimodal Learning

Many multimodal methods employing a joint training strategy struggle to fully exploit all modalities, resulting in under-optimized unimodal representations. Consequently, the performance of multimodal models often falls short of expectations, even though they may occasionally outperform their unimodal counterparts [17], [19].

To address this issue, Du et al. [18] tried to improve unimodal performance by distilling knowledge from well-trained models. However, these approaches introduce more model structure and computational effort, which makes the training process more complex and expensive. Peng et al. [19] chose to decrease the gradient of the dominant modality to lessen its inhibitory effect on the other modality. Xu et al. [21] introduced a modality-wise $L2$ normalization to features and weights toward balanced and better multimodal fine-grained learning. Although a certain degree of improvement is achieved, such approaches do not impose the intrinsic motivation of improvement on the slow-learning modality. Thus, Fan et al. [20] introduced the prototypical modal rebalance strategy to accelerate the slow-learning modality and alleviate the suppression of the dominant modality. Although these methods offer improvements to some extent, they predominantly focus on the weaker modality, overlooking the fact that the performance of the dominant modality in the multimodal model is also inferior to that in the unimodal model.

III. Methods

A. Overview

We propose a general framework GDNet to assist multimodal RS classification. As shown in Fig. 1, GDNet consists of two parts: gradient decoupled learning and unimodal regularization. For a specific, gradient decoupled learning guides each modality encoder to learn a probabilistic representation instead of a deterministic one. Consequently, the gradient for each modality encoder samples from the gradient distribution, alleviating their dependencies on the gradient passed from the shared fusion module. This decouples the representation learning of each modality encoder, promoting their optimization, including the dominant one. Unimodal regularization also contains two parts. The first part calculates task loss for each modality encoder, which introduces independent gradient paths for each modality encoder to accelerate their optimization and promote gradient decoupling. The second part aligns the logit output of each modality encoder with the multimodal one, encouraging them to preserve the modality-share information.

B. Notation

$\Theta = \{x_i, y_i\}_{i=1}^{N}$ is a multimodal dataset that contains N samples. Each x_i consists of V modalities $(m_1, \ldots, m_V)$, denoted as $x_i = (x_i^{m_1}, \ldots, x_i^{m_V})$. $E_v(.)$ denotes the modality encoder for the vth modality input. Without loss of generality, we assume $V = 2$ for simplicity. $\mu_i^{m_v}$ and ${\sigma_i}^{m_v}$ denote the mean and standard deviation for the output of the vth modality encoder. ${s_i}^{m_v}$ is the sampled

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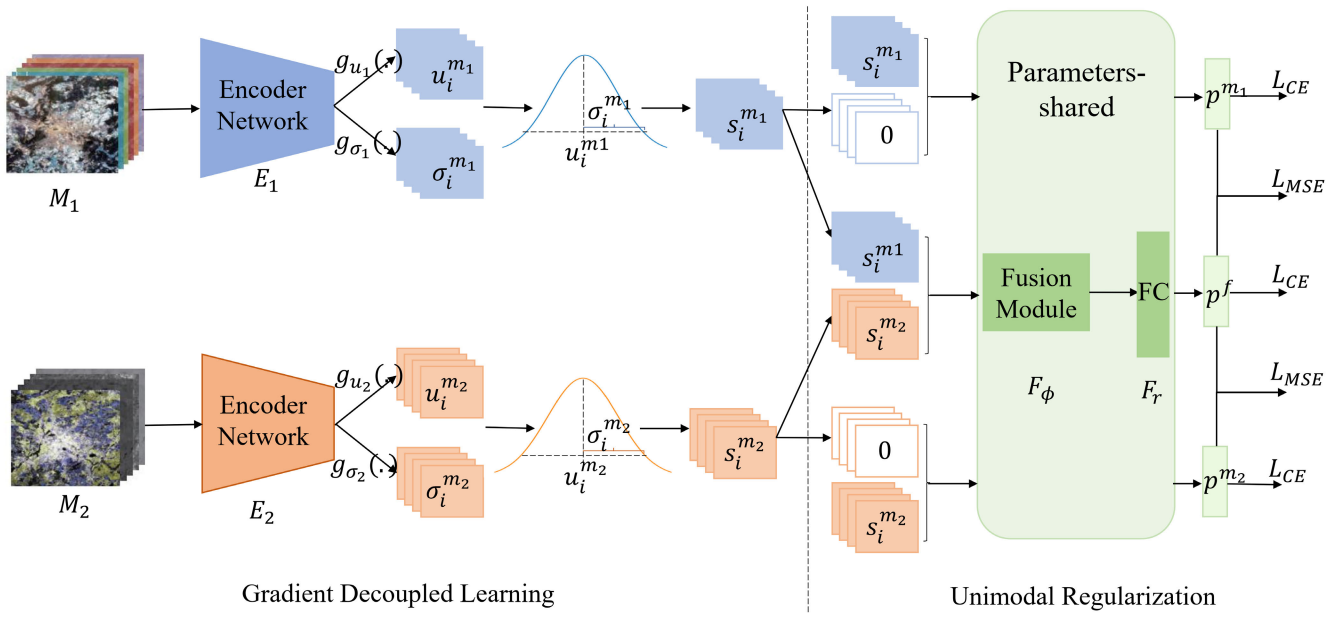


Fig. 1. Overall architecture of the proposed GDNet. It consists of two parts: gradient decoupling learning that guides each modality to learn probabilistic representation to alleviate their dependencies and the parameter-share unimodal regularization that guides each modality encoder to capture the task loss and align their logit output with the multimodal one, accelerating their optimization and enabling them to capture modality-share information. "FC" means the fully connected layer. Notations are described in Section III-B.

representation for the vth modality encoder. $F_\phi(.)$ and $F_r(.)$ are the map function of the fusion module and fully connected layer, respectively. p^{m_1}, p^{m_2}, and p^f are the logit output of m_1, m_2, and their fusing representation, respectively.

C. Gradient Decoupled Learning

The gradients for each modality encoder are influenced by the gradient of the shared fusion module and classifier in the multimodal model according to the chain rule. As a result, the optimization of each modality encoder is coupled with the others, resulting in insufficient learning. For example, the modality with better performance will dominate the optimization processing and converge quickly, decreasing the gradient of the shared modules and limiting the learning of weak modalities [19], [20]. To address this issue, this section introduces the gradient decoupled learning that guides each modality encoder to learn a probabilistic instead of a fixed representation. This converts the gradient of each modality encoder into a distribution, alleviating their dependencies on the gradient passed from the shared fusion module and decoupling their representation learning. Compared with existing methods, this can promote the optimization of not only the weak modality but also the dominant one.

Here, we take $x_i^{m_1}$ as an example. As shown in Fig. 1, given a feature map $E_1(x_i^{m_1})$ from the modality encoder, we model its probabilistic $s_i{}^{m_1}$ as a Gaussian distribution with mean $\mu_i{}^{m_1}$ and variance $(\sigma_i{}^{m_1})^2$

$$s_i{}^{m_1} \sim \mathcal{N}(s_i{}^{m_1}; \mu_i{}^{m_1}, (\sigma_i{}^{m_1})^2) \tag{1}$$

where both parameters $(\mu_i^{m_1}, (\sigma_i^{m_1})^2)$ of the Gaussian distribution are input-dependent and predicted by CNNs

$$\mu_i^{m_1} = g_{\mu_1}\left(\theta_u^{m_1}, E_1\left(x_i^{m_1}\right)\right) \tag{2}$$

$$\log\left(\left(\sigma_i{}^{m_1}\right)^2\right) = g_{\sigma_1}\left(\theta_\sigma^{m_1}, E_1\left(x_i^{m_1}\right)\right) \tag{3}$$

where $\theta_u^{m_1}$ and $\theta_\sigma^{m_1}$ refer to the parameters of the predictors g_{μ_1} and g_{σ_1}, respectively. Here, we calculate the logarithm of the variance instead of the actual variance for numerical stability [41]. In detail, we implement $g_{\mu_1}(\theta_u^{m_1}, \cdot)$ and $g_{\sigma_1}(\sigma_u^{m_1}, \cdot)$ with a simple 1×1 convolution with the batch norm, respectively, introducing negligible parameters.

Now, the representation of each sample becomes a stochastic embedding sampled from $\mathcal{N}(\mathbf{s}_i^{m_1}; \mu_i^{m_1}, (\sigma_i^{m_1})^2)$ in the latent space. However, the sampling operation is not differentiable, which prevents the backpropagation of the gradient flow during the model training. To this end, we use the reparameterization trick [42] to let the model still take gradients as usual. Specifically, we first sample a random noise ϵ from a normal distribution, which is independent of the model parameters, and then generate $s_i{}^{m_1}$ as the equivalent sampling representation

$$s_i{}^{m_1} = \mu_i^{m_1} + \epsilon\sigma_i^{m_1}, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{4}$$

Here, $s_i{}^{m_1}$ is the final representation of each image $x_i{}^{m_1}$. In the same way, we can get $s_i{}^{m_2}$ for $x_i{}^{m_2}$ as follows:

$$s_i{}^{m_2} = \mu_i^{m_2} + \epsilon\sigma_i^{m_2}, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{5}$$

Then, the multimodal logit output p^f is defined as follows:

$$p^f = F_r\left(F_\phi\left(s_i{}^{m_1}, s_i{}^{m_2}\right)\right). \tag{6}$$

And the classification loss L_{Multi} for the multimodal task at hand can be defined as follows:

$$L_{\text{Multi}} = L_{\text{CE}}\left(p^f, y_i\right) \tag{7}$$

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where L_{CE} is the cross-entropy loss, and the gradient for the encoder E_1 can be written as follows:

$$h_{E_1} = h_\tau \frac{\partial {\mu_i}^{m_1}}{\partial g_{\mu_1}} \frac{\partial g_{\mu_1}}{\partial E_1} + \epsilon h_\tau {\sigma_i}^{m_1} \frac{\partial f_{\sigma_1}}{\partial g_{\mu_1}} \frac{\partial g_{\mu_1}}{\partial E_1} \tag{8}$$

where h_τ is the gradient passed back from shared module, i.e.,

$$h_\tau = \frac{\partial L_{\mathrm{Multi}}}{\partial p^f} \frac{\partial p^f}{\partial F_r} \frac{\partial F_r}{\partial F_\phi} \frac{\partial F_\phi}{\partial {s_i}^{m_1}}. \tag{9}$$

Here, the gradient h_{E_1} also becomes a Gaussian distribution. Compared with the fixed gradient in the conventional multimodal model, the probabilistic gradient is sampled from a Gaussian distribution with mean and variance as $h_\tau(\partial {\mu_i}^{m_1}/\partial g_{\mu_1})(\partial g_{\mu_1}/\partial E_1)$ and $h_\tau {\sigma_i}^{m_1}(\partial f_{\sigma_1}/\partial g_{\mu_1})(\partial g_{\mu_1}/\partial E_1)$, respectively. This alleviates the explicit dependency of h_{E_1} and h_{E_2} on the shared gradient h_τ. In particular, their variance ${\sigma_i}^{h_1} = {\sigma_i}^{m_1}(\partial f_{\sigma_1}/\partial g_{\mu_1})(\partial g_{\mu_1}/\partial E_1)$ and ${\sigma_i}^{h_2} = {\sigma_i}^{m_2}(\partial f_{\sigma_1}/\partial g_{\mu_2})(\partial g_{\mu_2}/\partial E_2)$ control the level of relaxation for each modality encoder, respectively. A bigger variance will lead to a bigger sample range and result in a lower dependency.

Besides, previous studies show that the probabilistic model tends to learn the variance as zero to reduce the uncertainty for representation [43], [44], [45]. This reduction to zero diminishes the level of gradient decoupling. To this end, we introduce a regularization term to prevent the learned variances from collapsing to zero by introducing the KL divergence loss between the learned distributions and the standard normal $\mathcal{N}(\epsilon \mid \mathbf{0}, \mathbf{I})$

$$\begin{aligned} {L_{\mathrm{KL}}}^{m_1} &= KL\left[\mathcal{N}\left(\mathbf{s}_i^{m_1} | \mu_i^{m_1}, \left(\sigma_i^{m_1}\right)^2\right) \| \mathcal{N}(\epsilon \mid \mathbf{0}, \mathbf{I})\right] \\ &= -\frac{1}{2}\left(1 + \log\left(\sigma_i^{m_1}\right)^2 - \left(\mu_i^{m_1}\right)^2 - \left(\sigma_i^{m_1}\right)^2\right) \end{aligned} \tag{10}$$

where KL(.||.) means the KL divergence. Similarly, the regularization term for modality m_2 can be defined as follows:

$${L_{\mathrm{KL}}}^{m_2} = -\frac{1}{2}\left(1 + \log\left(\sigma_i^{m_2}\right)^2 - \left(\mu_i^{m_2}\right)^2 - \left(\sigma_i^{m_2}\right)^2\right). \tag{11}$$

And the total distribution regularization loss L_{DR} can be defined as follows:

$$L_{\mathrm{DR}} = {L_{\mathrm{KL}}}^{m_1} + {L_{\mathrm{KL}}}^{m_2}. \tag{12}$$

D. Unimodal Regularization

The unimodal regularization for GDNet contains two parts, label learning loss and distribution learning loss. Label learning loss guides each modality encoder to classify the object by its representation. This introduces an extra gradient path for the modality encoder to increase the variance for gradient distribution, promoting gradient decoupling and guiding the modality encoder to capture more modality-specific information. Distribution learning loss guides each modality encoder to align its logit output with the multimodal one, encouraging it to capture the modality-share information. More importantly, as shown in Fig. 1, the unimodal regularization shares fusion module and FC layers with the conventional multimodal model, introducing no extra complexity.

Here, we also take $x_i^{m_1}$ as an example. Its logit output p^{m_1} can be calculated by setting the representation of $x_i^{m_2}$ in (6) as 0

$$p^{m_1} = F_r\left(F_\phi\left({s_i}^{m_1}, 0 * {s_i}^{m_2}\right)\right) \tag{13}$$

and its label learning loss ${L_l}^{m_1}$ is defined as follows:

$${L_l}^{m_1} = L_{\mathrm{CE}}\left(p^{m_1}, y_i\right). \tag{14}$$

With the supervision of ${L_l}^{m_1}$, the gradient for encoder modality E_1 can be rewritten as follows:

$$\begin{aligned} h_{E_1} = &\left(h_\tau + {h_l}^{m_1}\right) \frac{\partial {\mu_i}^{m_1}}{\partial g_{\mu_1}} \frac{\partial g_{\mu_1}}{\partial E_1} \\ &+ \epsilon\left(h_\tau + {h_l}^{m_1}\right) {\sigma_i}^{m_1} \frac{\partial f_{\sigma_1}}{\partial g_{\mu_1}} \frac{\partial g_{\mu_1}}{\partial E_1} \end{aligned} \tag{15}$$

where ${h_l}^{m_1}$ is the gradient passed back from ${L_l}^{m_1}$. On the one hand, ${h_l}^{m_1}$ increases the variance of gradient distribution, enhancing the level of gradient decoupling and guiding the modality encoder to capture more modality-specific information. On the other hand, ${h_l}^{m_1}$ increases the mean gradient, accelerating the optimization of the modality encoder.

Besides, the distribution learning loss ${L_d}^{m_1}$ is defined as follows:

$${L_d}^{m_1} = L_{\mathrm{MSE}}\left(p^{m_1}, p^f\right) \tag{16}$$

where L_{MSE} denote the mean squared loss. To minimize this loss means minimizing the distance between the logit output of the fused representation and the unimodal representation. Since the fusion module and FC are shared, this loss will guide the model to capture the modality-share information.

Similarly, the label learning loss for modality m_2 is defined as follows:

$${L_l}^{m_2} = L_{\mathrm{CE}}\left(p^{m_2}, y_i\right) \tag{17}$$

and the distribution learning loss for modality m_2 is defined as follows:

$${L_d}^{m_2} = L_{\mathrm{MSE}}\left(p^{m_2}, p^f\right) \tag{18}$$

and the unimodal regularization loss L_{UR} can be defined as follows:

$$L_{\mathrm{UR}} = {L_l}^{m_1} + {L_l}^{m_2} + {L_d}^{m_1} + {L_d}^{m_2}. \tag{19}$$

E. Total Loss

Accordingly, the total loss L_{GDN} for the training of GDNet is defined as follows:

$$L_{\mathrm{GDN}} = L_{\mathrm{Multi}} + \alpha L_{\mathrm{DR}} + \beta L_{\mathrm{UR}} \tag{20}$$

here L_{Multi} is the classification loss for the multimodal task at hand. L_{DR} is the distribution regularization loss to prevent the learned variances from collapsing to zero. L_{UR} is the unimodal regularization loss to promote gradient decoupling and accelerate model optimization. α and β are the hyperparameters.

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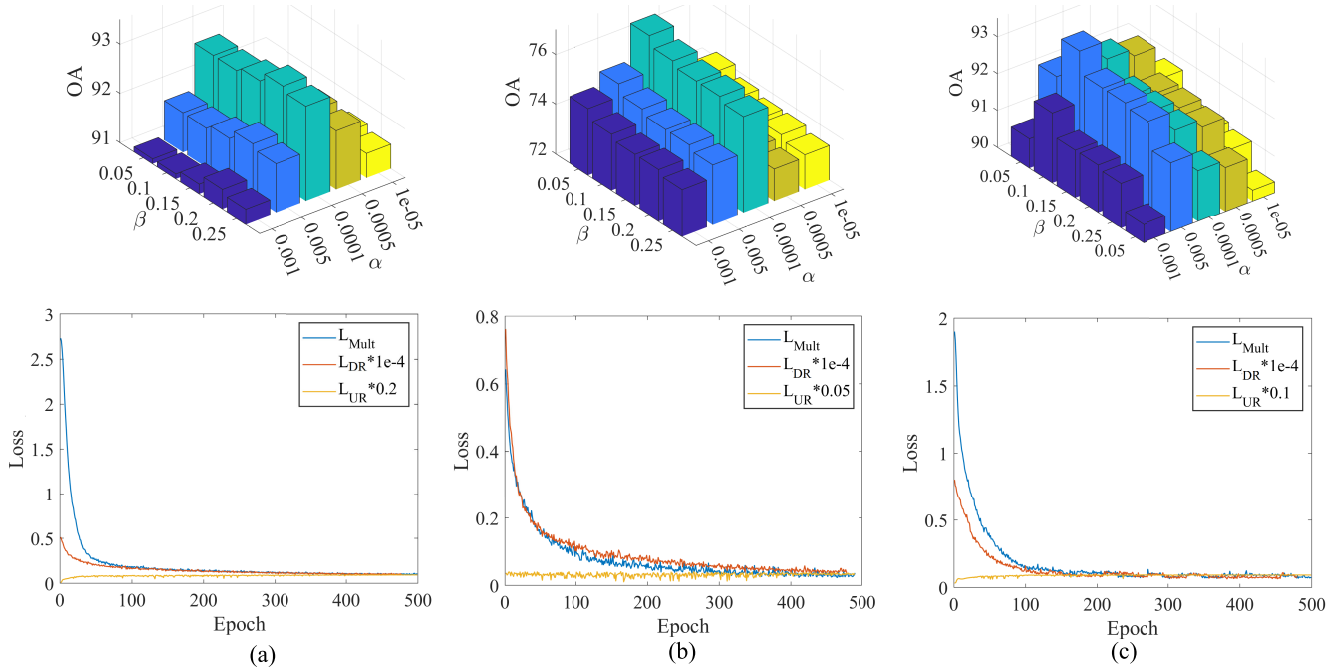


Fig. 2. Parameter analysis of GDNet on Huston2013 HS-LiDAR, Berlin HS- synthetic aperture radar (SAR), and Augsburg HS-digital surface model (DSM), respectively. The top line describes the trend between the overall accuracy (OA) and the parameters α, β on three modality combinations. The bottom line shows the loss curve of L_{Multi}, αL_{DR}, and βL_{UR}. (a) Huston2013 HS-LiDAR. (b) Berlin HS-SAR. (c) Augsburg HS-DSM.

IV. Experiments

A. Datasets

1) Huston2013 HS-LiDAR Dataset: This dataset contains one airborne HS and LiDAR data with dimensions of 349 × 1905 pixels [46]. Specifically, the HS image consists of 144 HS bands with a spatial resolution of 2.5 m. The corresponding LiDAR image is a grayscale image of the same spatial size as the HS image. The dataset encompasses 15 ground object classes, containing a total of 15 029 ground truth samples. Out of these, 2832 samples are allocated for training, and 12 197 for testing.

2) Berlin HS-SAR Dataset: This dataset is comprised of the simulated HS image and the SAR data that comes from Sentinel-1 payload [46]. Specifically, the HS image consists of 244 bands, with pixel dimensions of 797 × 220 pixels. The SAR data encompass four channels with identical pixel dimensions as the HS images. The dataset encompasses eight classes with a total of 464 671 pixels. For training and testing purposes, 2820 pixels are utilized for training, and 461 851 for testing.

3) Augsburg Dataset: This dataset consists of three different data sources, including a HS image, an SAR image, and a DSM image [46]. Specifically, the HS image is composed of 180 bands, the SAR image consists of four feature channels, and the DSM image is a grayscale image. Each band has dimensions of 332 × 485 pixels. The dataset encompasses seven different classes, and ground truth information is provided for a total of 78 294 pixels. Out of these, 761 pixels are designated for training, and 77 533 for testing.

B. Experiments Setup

1) Evaluation Metrics: In the experiments, we choose the multimodal land cover classification task to evaluate the proposed framework. This has been regarded as a potential application to evaluate the quality of the multimodal learning [46]. The evaluation use three standard metrics [29]: OA, Average Accuracy (AA), and Kappa Coefficient (Ka). These metrics provide a straightforward basis for quantitative performance comparison.

2) Comparison Methods: To study the effectiveness of the proposed GDNet, we have performed extensive experiments and compared it with state-of-the-art methods. The compared methods include shallow-based methods, i.e., CoSpace [24], LeMA [47], CNN-based methods, i.e., MDL_RS [29], CoupleNet [34], CCR-Net [5], CALC [48], as well as Transformer-based methods, i.e., MFT [15], DHViT [13], and ExViT [12].

3) Implementation Details: The proposed GDNet is designed to be versatile, and independent of the modality encoder and fusion method. It can seamlessly integrate into existing multimodal learning frameworks. Here, we implement it following the basic implementation of MDL-RS [29], which is a classic framework for multimodal RS classification.

The experiments were conducted on a computer equipped with a Ryzen 3600X processor, 64 GB RAM, and an RTX3090Ti-24 GB GPU. The entire framework was trained using the SGD optimizer with parameters set as follows: 500 training epochs, a batch size of 32, a learning rate of $1e^{-3}$, momentum of 0.9, and weight decay of $5e^{-3}$. Additionally, we employ the classic cosine learning rate policy for all models to dynamically adjust the learning rate at each iteration, enhancing overall performance. Finally, to handle the

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TABLE I
PERFORMANCE OF DIFFERENT COMPARISON METHODS ON THE HUSTON2013 HS-LIDAR DATASET IN TERMS OF AA, OA, AND KA. THE BEST ONES ARE BOLDED

Methods	CoSpace	LeMA	MDL-RS	CoupleNet	CCR-Net	CALC	MFT	DHViT	ExViT	GDNet
Healthy Grass	81.96	81.86	83.10	83.10	83.00	82.24	82.53	81.01	**91.55**	87.27
Stressed Grass	83.27	83.08	81.58	84.87	84.87	83.93	85.03	85.15	85.15	**91.02**
Synthetic Grass	**100.00**	**100.00**	**100.00**	99.80	**100.00**	93.47	98.55	93.47	98.61	99.20
Tree	94.22	94.79	**99.72**	92.42	92.14	98.86	94.60	80.40	98.61	97.30
Soil	99.34	99.34	99.81	99.24	99.81	99.72	99.84	99.53	**100.00**	99.81
Water	**99.30**	99.30	95.10	95.80	95.80	98.60	95.10	95.80	98.60	95.80
Residential	81 .44	88.99	90.02	95.24	**95.34**	90.21	85.79	92.16	88.90	90.29
Commercial	66.10	74.26	87.94	81.86	81.39	81.58	79.87	92.21	**93.35**	87.84
Road	69.97	73.84	81.59	85.08	84.14	84.99	89.05	81.78	**89.52**	87.20
Highway	48.94	72.20	86.68	61.10	63.22	68.15	62.45	68.73	65.54	**88.94**
Railway	88.61	82.26	89.37	83.87	90.32	95.16	98.04	79.60	95.83	**98.05**
Parking Lot	88.57	90.30	85.69	91.26	93.08	93.56	95.84	94.81	90.39	**97.93**
Parking Lot2	68.07	67.37	83.16	88.77	88.42	87.37	91.81	88.42	**90.53**	85.45
Tennis Court	**100.00**	**100.00**	**100.00**	91.09	96.36	99.60	99.73	**100.00**	99.60	98.58
Running Track	98.31	98.10	98.73	98.73	99.37	99.58	92.60	71.25	89.85	**99.47**
OA	82.14	85.42	89.60	87.23	88.15	90.61	88.45	85.82	87.62	**92.63**
AA	84.54	87.05	90.83	88.82	89.82	90.97	90.05	86.95	89.16	**92.91**
Ka	0.8062	0.8417	0.8875	0.8619	0.8719	0.8918	0.8746	0.8461	0.8683	**0.9143**

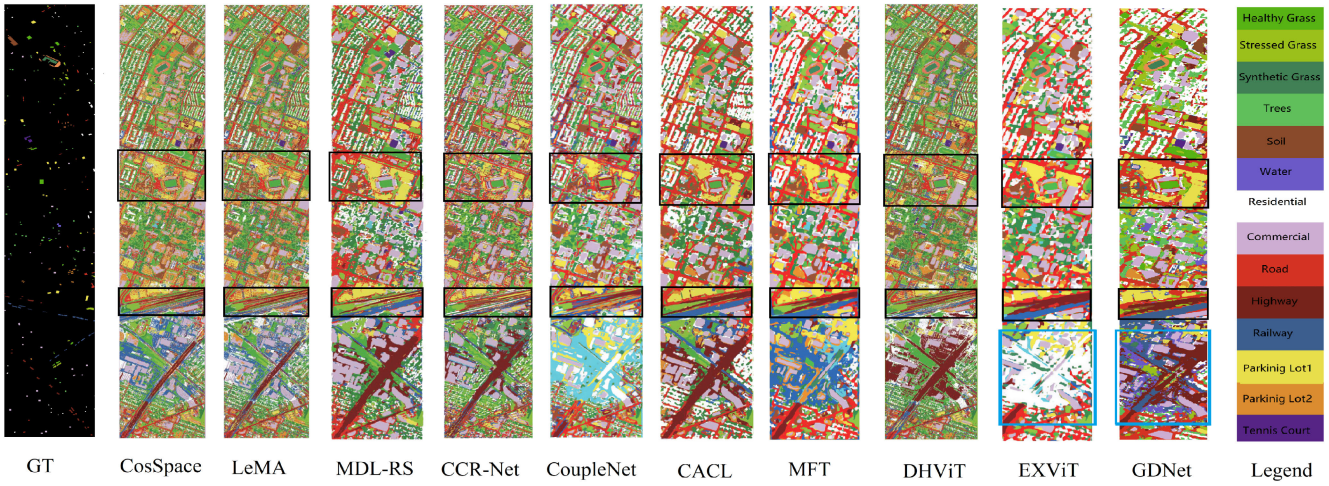


Fig. 3. Classification maps of different compared algorithms on Huston2013 HS-LiDAR dataset.

randomness in initialization, each experiment was performed independently five times, and the mean value was selected as the final result.

Data preparation is crucial for performance. In contrast to conventional classification tasks where the entire image is the input, RS image classification involves cropping a patch centered around a pixel. To ensure a fair comparison, we adopted the patch cropping method from previous work [29], extending the original pixel using the "replicate" operation, with a patch size set to 7×7. Additionally, random flipping was employed for data augmentation.

C. Configuration of Hyperparameters

The proposed GDNet has two hyperparameters: α for distribution regularization and β for unimodal regularization, which would influence the land cover classification performance. Therefore, we conduct experiments on three datasets to search for the optimal hyperparameters. According to the value of L_{Multi}, L_{DR}, and L_{UR}, we set the values of the parameters α and β for L_{DR} range in $(0.05, 0.1, 0.15, 0.2, 0.25)$ and $(1e^{-3}, 5e^{-3}, 1e^{-4}, 5e^{-4}, 1e^{-5})$, respectively, so that αL_{DR}, and βL_{UR} can be converted into the same magnitude as L_{Multi}. This has been shown effective for multitask learning [49].

In Fig. 2, the optimal (α, β) values for the Huston2013 HS-LiDAR, Berlin HS-SAR, and Augsburg HS-DSM datasets are determined to be $(1e^{-4}, 0.2)$,$(1e^{-4}, 0.05)$, and $(1e^{-4}, 0.1)$, respectively, as indicated in the top line. These coefficients effectively regulate the three losses to fall within the same range. Additionally, the top line of Fig. 2 illustrates that excessively large or small values of α can detrimentally impact model performance across all datasets. An overly large α may lead to convergence issues, as the estimated distribution becomes a completely random Gaussian distribution. Conversely, an excessively small α diminishes the effectiveness of distribution regularization, constraining the sampling range of the probabilistic representation. Consequently, for the Augsburg HS-SAR dataset, the chosen (α, β) values are set as $(1e^{-4}, 0.1)$.

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TABLE II
PERFORMANCE OF DIFFERENT COMPARISON METHODS ON THE BERLIN HS-SAR DATASET IN TERMS OF AA, OA, AND KA. THE BEST ONES ARE BOLDED

Methods	CoSpace	LeMA	MDL-RS	CoupleNet	CCR-Net	CALC	MFT	DHViT	ExViT	GDNet
Forest	85.09	85.11	79.31	84.09	85.93	65.15	67.12	72.41	70.44	**79.69**
Residential Area	61.60	64.84	70.46	68.48	71.04	71.36	71.16	71.11	72.48	**85.54**
Industrial Area	51.18	48.94	41.64	49.09	41.80	42.02	40.45	29.43	46.95	**54.04**
Low Plants	75.44	80.04	85.82	79.43	82.62	84.57	88.54	88.69	81.71	74.73
Soil	82.50	80.66	84.94	81.25	85.10	92.67	85.22	80.58	91.35	**92.02**
Allotment	54.66	54.07	61.01	50.68	63.02	68.57	63.12	66.56	79.16	38.50
Commercial Area	28.81	27.40	37.84	26.16	29.23	37.30	36.93	35.22	36.35	28.8
Water	60.78	57.75	76.98	59.52	68.78	77.52	76.62	73.50	79.81	72.91
OA	64.53	66.71	70.82	68.51	69.85	71.44	70.66	70.14	70.01	**76.87**
AA	62.51	62.35	68.50	62.34	66.99	**67.39**	67.12	64.69	66.78	65.42
Ka	0.5093	0.5312	0.5808	0.5476	0.5716	0.5853	0.5839	0.5735	0.5834	**0.6632**

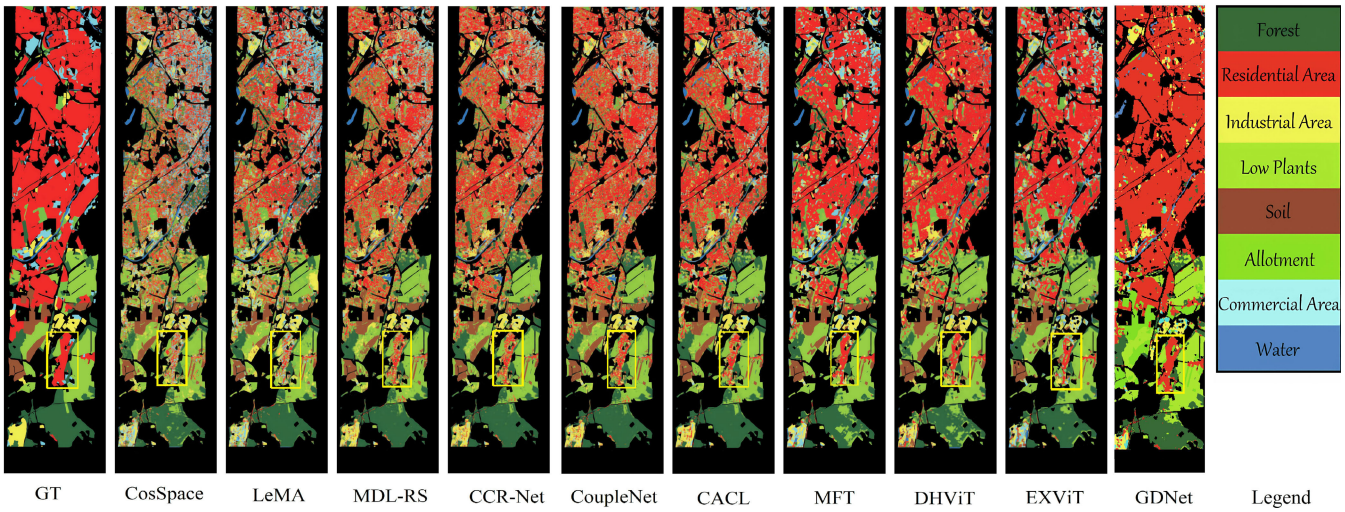


Fig. 4. Classification maps of different compared algorithms on Berlin HS-SAR dataset.

D. Comparison Result and Analysis

In this section, we will compare the performance of GDNet and its competitors quantitatively and qualitatively.

1) Quantitative Results and Analysis: Tables I–IV clearly show the OA, AA, and Ka for both the proposed method and each comparative method across the Houston2013, Berlin, and Augsburg datasets, respectively. In each table, the best results are highlighted in bold. In general, the values of the evaluation indicators consistently demonstrate that the proposed GDNet performs the best and is significantly more accurate than other methods.

Tables I and III present the results for the modality combination of HS and LiDAR as well as HS and DSM, respectively. As shown in Table I, the proposed GDNet achieves the performance of 92.63%, 92.91%, and 0.9143 in terms of the OA, AA, and Ka, respectively. This outperforms the second-best method, CALC, by 1.92%, 1.94%, and 2.30%, respectively. More importantly, its accuracy in all categories is above 85%. These results demonstrate the effectiveness of decoupling the learning of different modality encoders and accelerating their learning independently. Besides, as shown in Table III, GDNet outperforms the second-best method by 1.87%, 6.52%, and 2.06% in terms of the OA, AA, and Ka in the Augsburg dataset when using the HS and DSM modalities. Moreover, it also achieves the best performance in all classes except for the commercial area. These results verify its effectiveness in dealing with HS and LiDAR or DSM modalities.

Tables II and IV show the results on the modality combination of HS and SAR. First, in Table II, compared with the second-best result, GDNet increases the OA from 71.44% to 76.87% in the Berlin dataset, bringing a significant improvement. Besides, Table IV shows that GDNet outperforms the second-best method by 2.05%, 8.36%, and 3.05% in terms of the OA, AA, and Ka, respectively, in the Augsburg dataset when using the HS and SAR modalities. It achieves the best performance in all classes except for the commercial area. These results demonstrate that the proposed GDNet is not only suitable for HS and LiDAR modalities but also works with other modalities, such as SAR.

2) Visual Evaluation and Analysis: Figs. 3–5 show the classification maps of GDNet and its comparison methods on Huston2013 HS-LiDAR, Berlin HS-SAR, and Augsburg HS-DSM datasets, respectively. It can be seen from the maps that the proposed method produces a better classification performance with clearer boundaries when compared with the other methods, which is consistent with the numerical results.

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TABLE III
PERFORMANCE OF DIFFERENT COMPARISON METHODS ON THE AUGSBURG DATASET WITH HS AND DSM MODALITIES IN TERMS OF AA, OA, AND KA. THE BEST ONES ARE BOLDED

Methods	CoSpace	LeMA	MDL-RS	CoupleNet	CCR-Net	CALC	MFT	DHViT	ExViT	GDNet
Forest	82.04	78.85	**98.82**	95.29	93.47	94.16	94.77	90.45	91.83	97.06
Residential Area	83.52	95.12	94.96	96.51	96.86	95.32	96.03	90.87	95.38	**97.87**
Industrial Area	34.99	61.98	71.44	68.59	82.56	**86.18**	72.52	61.20	43.32	83.31
Low Plants	76.77	76.77	92.25	82.43	84.45	95.57	93.52	82.82	91.13	**97.19**
Allotment	52.77	20.49	**69.41**	75.52	44.36	10.00	51.31	21.80	41.11	56.02
Commercial Area	14.22	9.46	7.39	15.99	0.00	6.05	13.49	23.50	26.01	0.0
Water	46.71	46.78	43.33	48.37	40.48	45.96	31.28	7.43	42.07	**48.50**
OA	76.17	78.25	89.13	87.31	88.85	91.46	90.49	86.06	89.82	**93.33**
AA	56.13	52.29	67.08	68.96	66.99	61.89	64.70	54.01	61.41	**68.41**
Ka	0.6643	0.6811	0.8451	0.8205	0.8316	0.8804	0.8635	0.8591	0.8644	**0.9011**

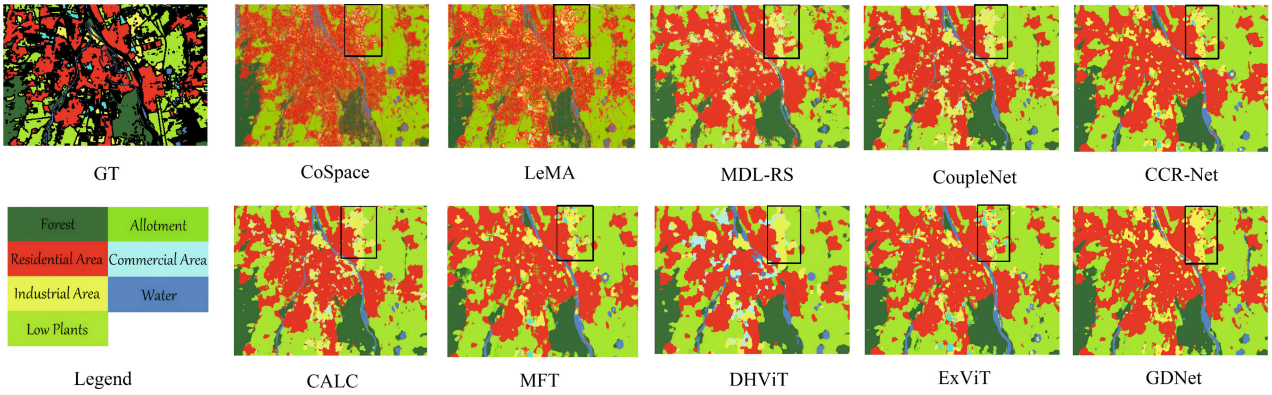


Fig. 5. Classification maps of GDNet and its compared algorithms on the Augsburg dataset with HS and DSM modalities.

As shown in Fig. 3, GDNet generates more complete and accurate predictions on the parking lot1 (in the top black box) and highway (in the bottom black box). In contrast, the classification maps of other competitors are intermittent. This is consistent with the numerical results in Table I, where GDNet achieves the best performance in both the parking lot1 and the highway. Similarly, the visualizations of GDNet on Berlin HS-SAR, and Augsburg HS-DSM datasets are also consistent with the numerical results. GDNet produces better classification performances in the residential area (see the yellow block in Fig. 4) and industrial area (see the black block in Fig. 5) and has clearer boundaries when compared with the other methods.

It is worth noting that some visual results do not appear to align with their quantitative performance. Take the highway type in the Huston HS-LiDAR dataset as an example, the quantitative performance of GDNet and ExViT on it are 88.90 and 65.54. However, their visual performance in the black block in Fig. 3 has similar results. This is because the test set is only a part of the whole image. Specifically, the Huston2013 HS-LiDAR dataset contains 664 845 (349 × 1905) patches, of which 12 197 patches are used for testing, accounting for only 1.83% of all patches. Thus, the quantitative results from a small number of patches may be mismatched with the visualization results of the whole image, including the highway type.

E. Ablation Study

In this section, we first study the effectiveness of GDNet to promote the optimization of the unimodal branch. Then, we study the generalization of GDNet to different frameworks, including the CNN-based and Transformer-based ones, respectively. Finally, we study the effect of patch size. To this end, we conduct extensive ablation experiments on the Huston2013 HS-LiDAR dataset.

1) Effect to Promote the Unimodal Branch Optimization: To evaluate the optimization of unimodal branches, we report the performance of each unimodal branch in the multimodal model through the unimodal prediction calculated via 13 on the Huston2013 HS-LiDAR dataset. The results are shown in Table V. "Vanilla" denotes the baseline multimodal model, which is implemented with the MDL-RS model [29]. "GB" denotes the gradient-blending method that improves unimodal learning in multimodal learning via auxiliary classifiers [17]. "PMR" denotes the prototype modality rebalanced method that assists the under-optimized multimodal model learning via prototype learning.

As shown in Table V, the HS branch of the vanilla model outperforms the LiDAR branch by 48.42% in terms of OA. This demonstrates that the vanilla multimodal model pays more attention to the HS modality while ignoring the LiDAR modality. Consequently, we need to improve unimodal learning in the multimodal RS model. Besides, all balanced methods improve the performance of the LiDAR branch, verifying their effectiveness in improving unimodal learning in multimodal learning. Among them, GDNet achieves the best results. Specifically, compared with the vanilla model, GDNet improves the OA LiDAR branch by 32.37%, respectively. Besides, while the conventional balanced methods can improve the performance of the LiDAR branch, they decrease the

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TABLE IV

PERFORMANCE OF DIFFERENT COMPARISON METHODS ON THE AUGSBURG DATASET WITH HS AND SAR MODALITIES IN TERMS OF AA, OA, AND KA. THE BEST ONES ARE BOLDED

Methods	CoSpace	LeMA	MDL-RS	CoupleNet	CCR-Net	CALC	MFT	DHViT	ExViT	GDNet
Forest	90.18	90.55	97.52	96.28	96.72	90.60	94.65	90.01	92.28	**98.07**
Residential Area	87.41	89.81	93.91	96.50	96.93	98.43	**96.90**	94.27	96.53	96.58
Industrial Area	20.59	23.03	70.84	63.42	48.93	46.14	69.80	64.62	65.53	**89.32**
Low Plants	76.16	83.73	91.77	85.05	88.62	**97.47**	93.98	85.53	90.79	94.71
Allotment	28.16	34.23	10.81	44.30	46.29	49.08	32.70	29.00	49.46	**55.83**
Commercial Area	5.12	9.71	10.08	11.12	18.33	17.94	10.52	20.41	17.58	0.0
Water	38.03	45.92	42.21	46.91	**48.31**	47.47	23.98	44.17	44.43	46.31
OA	79.96	81.60	88.83	87.58	89.14	90.06	90.49	88.15	90.11	**92.54**
AA	49.09	53.85	59.04	63.36	63.44	63.87	60.36	61.14	65.22	**68.69**
Ka	0.6973	0.6973	0.8322	0.8214	0.8452	0.8504	0.8626	0.8488	0.8244	**0.8931**

TABLE V

UNIMODAL AND MULTIMODAL PERFORMANCE OF THE MODEL USING DIFFERENT BALANCED LEARNING STRATEGIES ON THE HUSTON2013 HS-LIDAR DATASET IN TERMS OF OA. ● DENOTES THE MODALITY REPRESENTATION IS AVAILABLE. ○ DENOTES THE MODALITY REPRESENTATION IS MISSING, I.E., SET AS 0

Metric	Modalities		Methods			
	HS	LiDAR	Vanilla	GB	PMR	GDNet
OA	●	○	67.58	65.42	57.89	**87.62**
	○	●	19.16	32.44	40.25	**51.82**
	●	●	89.60	89.91	90.51	**92.63**
AA	●	○	71.61	70.01	62.12	**88.90**
	○	●	14.75	29.34	38.55	**54.89**
	●	●	90.83	90.32	90.91	**92.91**
Ka	●	○	0.6481	0.6585	0.7102	**0.8659**
	○	●	0.1155	0.2677	0.3568	**0.4832**
	●	●	0.8875	0.8943	0.9001	**0.9143**

TABLE VI

PERFORMANCE OF DIFFERENT MULTIMODAL FRAMEWORKS WHEN USING AND NOT USING GDNET

Metric	Framework			
	CCRNet		ExViT	
	Vanilla	GDNet	Vanilla	GDNet
OA	88.15	**91.60**	87.62	**88.73**
AA	89.82	**92.11**	89.16	**90.04**
Ka	0.8719	**0.8991**	0.8683	**0.8782**

performance of the HS branch due to the coupled gradient. In contrast, GDNet decouples the gradient of each branch, improving the performance of HS and LiDAR branches simultaneously.

2) Generalization to Different Frameworks: Here, we report and compare the performance of GDNet on the CCR-Net and ExViT, respectively, to study its generalization ability to different frameworks. The results are shown in Table VI. Here, "Vanilla" denotes the baseline methods that do not use GDNet.

As shown in Table VI, DGDNet improves the OA, AA, and Ka of CCR-Net by 3.45% 2.29%, and 2.72%, respectively. Besides, ExViT with GDNet also outperforms that without GDNet by 1.11%, 0.88%, and 0.99%, respectively. These results demonstrate that GDNet can be integrated into other multimodal RS models and improve their performance.

3) Effect of Patch Size: We report and compare the performance of the proposed GDNet implemented with different patch sizes. The results are shown in Fig. 6. From

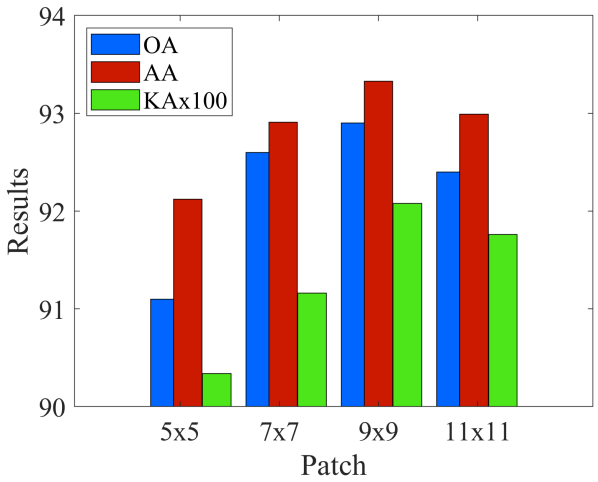


Fig. 6. Performance comparison of the GDNet implemented with different patch sizes.

this figure, we can see that the OA, AA, and Ka will gradually increase to the maximum value and then decrease as the patch increases. This is because bigger patches usually contain more information, and improve the model performance. However, due to the fixed structure and downsampling coefficients of the network, its ability to handle bigger input is insufficient. Consequently, the performance decreases when the input patch size is too big.

V. CONCLUSION AND LIMITATION

A. Conclusion

In this article, we analyze the imbalanced multimodal learning problem in multimodal RS image classification and introduce GDNet to overcome it. Specifically, GDNet comprises two key components: gradient decoupled learning and unimodal regularization. Gradient decoupled learning guides each modality to learn a probabilistic instead of fixed embedding. This helps transfer the deterministic gradients of each modality encoder into a distribution, decoupling their dependency on each other and promoting unimodal branch optimization. Besides, unimodal regularization guides each modality encoder to calculate the task loss independently and align their logit output with the multimodal one. This can accelerate the learning of each modality for modality-specific

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information when preserving the modality-share information. Importantly, both components are agnostic to the underlying backbone and fusion methods, making them adaptable to integrate with diverse multimodal RS models for performance enhancement. Experiments and ablation studies demonstrate the effectiveness and generalization of the proposed GDNet to promote multimodal RS image classification.

B. Limitation

Although GDNet has achieved promising results in multimodal RS land cover classification tasks, it only considers the imbalance problem from the modality level and ignores the effect of sample difference. However, the strong and weak modalities of different samples may be inconsistent. Besides, GDNet focuses on the modality imbalance problem in multimodal RS classification tasks, ignoring the modality imbalance problem in dense prediction tasks such as multimodal RS segmentation and detection, which also deserves further study. We leave these challenges in the future works.

REFERENCES

[1] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[2] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[3] W. Liao, R. Bellens, A. Pižurica, S. Gautama, and W. Philips, "Combining feature fusion and decision fusion for classification of hyperspectral and LiDAR data," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1241–1244.

[4] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[5] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2021.

[6] X. Li, L. Lei, and G. Kuang, "Multi-modal fusion architecture search for land cover classification using heterogeneous remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 5997–6000.

[7] X. Li, L. Lei, Y. Sun, M. Li, and G. Kuang, "Multimodal bilinear fusion network with second-order attention-based channel selection for land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1011–1026, 2020.

[8] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[9] X. Zhao, R. Tao, W. Li, W. Philips, and W. Liao, "Fractional Gabor convolutional network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.

[10] X. Li, L. Lei, Y. Sun, M. Li, and G. Kuang, "Collaborative attention-based heterogeneous gated fusion network for land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3829–3845, May 2021.

[11] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[12] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[13] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[14] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2022.

[15] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[16] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5500716.

[17] W. Wang, D. Tran, and M. Feiszli, "What makes training multi-modal classification networks hard?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12692–12702.

[18] C. Du et al., "Improving multi-modal learning with uni-modal teachers," 2021, *arXiv:2106.11059*.

[19] X. Peng, Y. Wei, A. Deng, D. Wang, and D. Hu, "Balanced multimodal learning via on-the-fly gradient modulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8228–8237.

[20] Y. Fan, W. Xu, H. Wang, J. Wang, and S. Guo, "PMR: Prototypical modal rebalance for multimodal learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20029–20038.

[21] R. Xu, R. Feng, S.-X. Zhang, and D. Hu, "MMCosine: Multi-modal cosine loss towards balanced audio-visual fine-grained learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[22] D. Tuia and G. Camps-Valls, "Kernel manifold alignment for domain adaptation," *PLoS ONE*, vol. 11, no. 2, Feb. 2016, Art. no. e0148655.

[23] L. Yan, M. Cui, and S. Prasad, "Joint Euclidean and angular distance-based embeddings for multisource image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1110–1114, Jul. 2018.

[24] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[25] D. Hong, J. Chanussot, N. Yokoya, J. Kang, and X. X. Zhu, "Learning-shared cross-modality representation using multispectral-LiDAR and hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1470–1474, Aug. 2020.

[26] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[27] S. Wei, Y. Luo, and C. Luo, "Diversity-guided distillation with modality-center regularization for robust multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4412114.

[28] S. Wei, Y. Luo, X. Ma, P. Ren, and C. Luo, "MSH-net: Modality-shared hallucination with joint adaptation distillation for remote sensing image classification using missing modalities," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402615.

[29] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.

[30] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[31] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, S. K. Roy, and X. Wu, "Hybrid dense network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observat. Remote Sens.*, vol. 15, pp. 3948–3957, 2022.

[32] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[33] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[34] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[35] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:28:00 UTC from IEEE Xplore. Restrictions apply.

[36] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2020.

[37] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based spectrospatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 92–93.

[38] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2022.

[39] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[40] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[41] S. Wei, Y. Luo, Y. Wang, and C. Luo, "Robust multimodal learning via representation decoupling," 2024, *arXiv:2407.04458*.

[42] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[43] Y. Shi and A. Jain, "Probabilistic face embeddings," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6902–6911.

[44] J. Chang, Z. Lan, C. Cheng, and Y. Wei, "Data uncertainty learning in face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5710–5719.

[45] S. Chun, S. J. Oh, R. S. De Rezende, Y. Kalantidis, and D. Larlus, "Probabilistic embeddings for cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8415–8424.

[46] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[47] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.

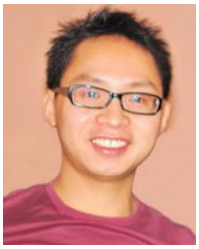
[48] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[49] S. Vandenhende, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Trans. pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3614–3633, Jan. 2021.



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